Class 10 Science: Chemistry

NCERT syllabus 2026-27 par based

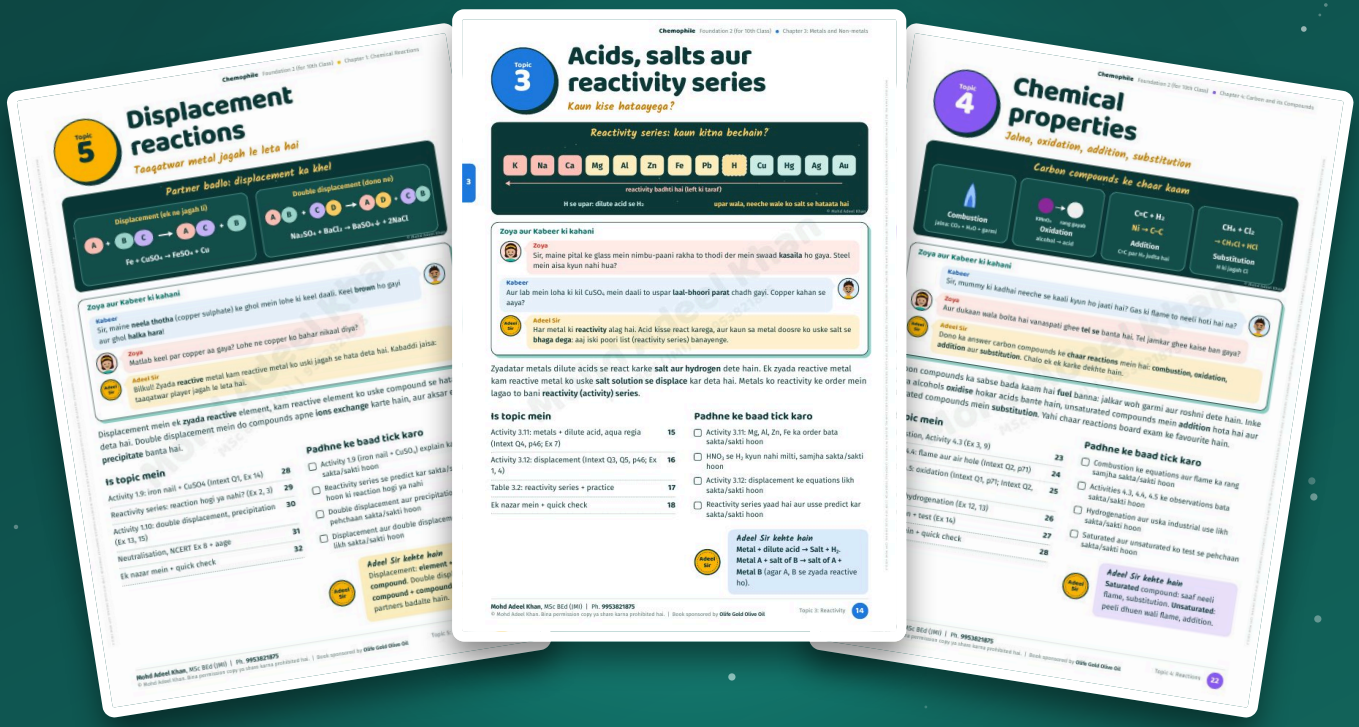
Chemophile

Foundation 2 (for 10th Class)

FREE SAMPLE
har chapter se ek topic

Har chapter ki ek jhalak

NCERT Class 10 Science ke saare 4 chemistry chapters, easy Hinglish mein, Zoya aur Kabeer ke saath



1 Chemical Reactions and Equations
2 Acids, Bases and Salts
3 Metals and Non-metals
4 Carbon and its Compounds

210 colourful pages | **124** figures aur diagrams | **51** solved examples | **500+** practice questions | **4** chapter tests (40 marks)

Adeel Sir

Mohd Adeel Khan
MSc BEd (JMI) | Ph. **9953821875**

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Chemophile kya hai?

Chemophile Foundation 2 (for 10th Class) NCERT Class 10 Chemistry ke saare **4 chapters** ki colourful booklets hain. Explanations **modern Hinglish** mein, technical terms aur exam answers **English** mein. Har topic ek story ki tarah chalta hai: Adeel Sir ka example, visual, concept, exam wala answer, aur end mein practice.

4	210	124	51	500+
chapters	colourful pages	figures aur diagrams	solved examples	practice questions

Har page par kya milega

Zoya aur Kabeer ki kahani	Har topic ek chhoti si story se shuru hota hai.
NCERT Activity	Activity kaise karein, kya dikhta hai, aur kyun.
Galti mat karna	Common mistakes jo marks cut karwati hain.
Mystery clue	Har chapter ek mystery; clues collect karke solve karo.
Adeel Sir ka example	Har concept ghar ki kisi cheez se samjhaya gaya hai.
Exam mein aise likho	Board exam ke liye English answer, ruled lines par.
Aage kaam aayega	Thoda extra jo Class 11 aur 12 mein kaam aayega.
Ghar par karo	Ghar ki cheezon se chhoti, safe activities.

Har chapter ke andar

- ✓ Har topic: Zoya-Kabeer ki story, example, visual, phir concept
- ✓ NCERT ke saare exercise questions solved
- ✓ Har topic ke end mein recap, quick check aur answers
- ✓ Har chapter ke end mein revision aur 40 marks ka test
- ✓ NCERT ki saari activities step by step
- ✓ Exam mein aise likho: board ke liye English answers
- ✓ Har chapter ka mystery, crossword aur Galti dhundo
- ✓ Bonus booklet, revision sheets aur mock tests alag se

Is sample mein kya hai?

Har chapter se **ek topic** ke do asli pages: topic ka pehla page aur uska pehla lesson page. Phir Chapter 2 ke revision ke do pages (fact sheet aur chapter test ka pehla page). Yeh pages bilkul waise hi hain jaise poori book mein: **210 pages** isi style mein.

Adeel Sir kehte hain

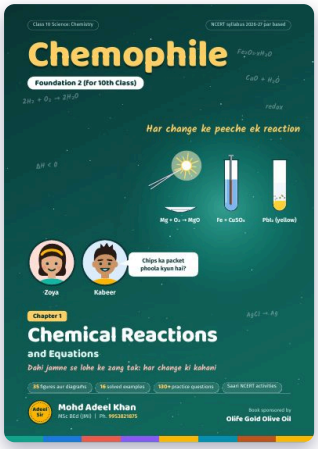
Chemistry ratne ka subject nahi, samajhne ka subject hai. Jab concept dikh jaata hai, to formula khud yaad ho jaata hai.

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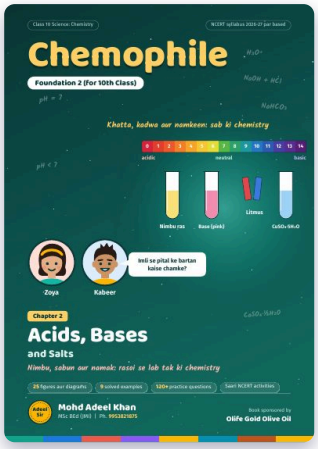
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Saare 4 chapters ek nazar mein

Har chapter ki apni booklet hai. Neeche har booklet ka cover, uske pages aur is sample mein shaamil topic.



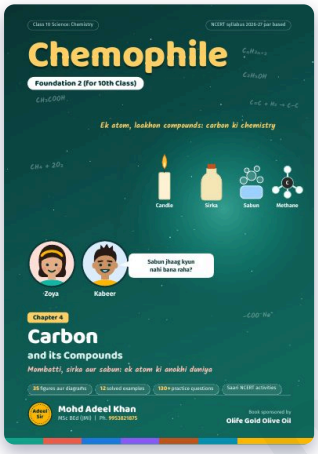
Chapter 1 Chemical Reactions and Equations
54 pages · Sample: Displacement aur double displacement



Chapter 2 Acids, Bases and Salts
49 pages · Sample: pH scale aur rozana zindagi



Chapter 3 Metals and Non-metals
51 pages · Sample: Acids, salt solutions aur reactivity series

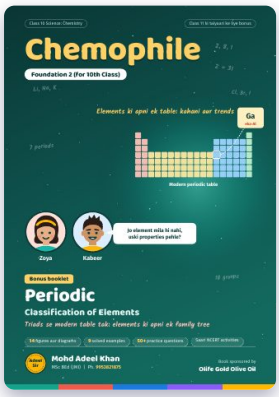


Chapter 4 Carbon and its Compounds
56 pages · Sample: Carbon compounds ki chemical properties

Poori series

4 chapters, 210 pages, har chapter ek alag booklet. Har booklet mein cover, contents, topics, revision aur chapter test.

Saath mein extra booklets



Bonus: Periodic Classification
29 pages · Class 11 ki taiyaari



Quick Revision Sheets
Har chapter ek page



Mock Test Papers
2 papers · answer key



Complete Volume
250 pages · sab ek saath

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Topic 5

Displacement reactions

Taaqatwar metal jagah le leta hai

Partner badlo: displacement ka khel

Displacement (ek ne jagah li)



$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Double displacement (dono ne)



$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$

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Zoya aur Kabeer ki kahani



Kabeer
Sir, maine **neela thotha** (copper sulphate) ke ghol mein lohe ki keel daali. Keel **brown** ho gayi aur ghol **halka hara**!



Zoya
Matlab keel par copper aa gaya? Lohe ne copper ko bahar nikaal diya?

Adeel Sir

Adeel Sir
Bilkul! Zyada **reactive** metal kam reactive metal ko uski jagah se hata deta hai. Kabaddi jaisa: taaqatwar player jagah le leta hai.

Displacement mein ek **zyada reactive** element, kam reactive element ko uske compound se hata deta hai. Double displacement mein do compounds apne **ions exchange** karte hain, aur aksar ek **precipitate** banta hai.

Is topic mein

Activity 1.9: iron nail + CuSO4 (Intext Q1, Ex 14)	28
Reactivity series: reaction hogi ya nahi? (Ex 2, 3)	29
Activity 1.10: double displacement, precipitation (Ex 13, 15)	30
Neutralisation, NCERT Ex 8 + aage	31
Ek nazar mein + quick check	32

Padhne ke baad tick karo

- ☐ Activity 1.9 (iron nail + $CuSO_4$) explain kar sakta/sakti hoon
- ☐ Reactivity series se predict kar sakta/sakti hoon ki reaction hogi ya nahi
- ☐ Double displacement aur precipitation reaction pehchaan sakta/sakti hoon
- ☐ Displacement aur double displacement ka farq likh sakta/sakti hoon

Adeel Sir

Adeel Sir kehte hain
Displacement: **element + compound**. Double displacement: **compound + compound**, dono ke partners badalte hain.

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Section 1.2.3

Samajh aaya? Rate karo ☆ ☆ ☆

Activity 1.9: lohe ki keel aur neela ghol

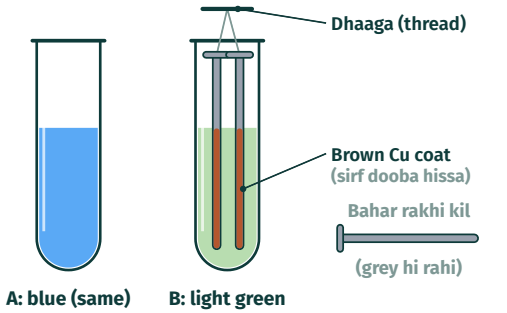
NCERT Activity 1.9 · iron nails in copper sulphate solution

Kya karna hai

1. 3 iron nails ko sandpaper se saaf karo; 2 test tubes (A, B) mein 10 mL $CuSO_4$ solution lo.
2. Do nails ko dhaage se baandh kar B mein 20 minute dubo do; ek nail comparison ke liye bahar rakho.
3. Nails nikaal kar solution ka colour (A vs B) aur nails ka colour compare karo.

Kya dikhta hai

B ka **blue colour halka** (light green) ho jaata hai aur dubi nails par **brown** (copper) layer aa jaati hai.



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$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$ (blue → light green)

Iron, copper se **zyada reactive** hai, isliye usne copper ko $CuSO_4$ se hata diya. Isi tarah zinc aur lead bhi copper ko hata dete hain:

$$Zn_{(s)} + CuSO_{4(aq)} \rightarrow ZnSO_{4(aq)} + Cu_{(s)}$$

$$Pb_{(s)} + CuCl_{2(aq)} \rightarrow PbCl_{2(aq)} + Cu_{(s)}$$

Displacement reaction: jab ek **zyada reactive** element, kam reactive element ko uske compound (solution) se bahar nikaal deta hai.

⚠ **Galti mat karna.** Nails ko pehle **sandpaper** se saaf karna zaroori hai: rust ya oil ki layer reaction ko rok deti hai. Aur nail par jo brown layer aati hai woh **copper** hai, rust nahi!

Exam mein aise likho

Intext Q1 (page 13). Why does the colour of copper sulphate solution change when an iron nail is dipped in it?

Iron is more reactive than copper. So it displaces copper from copper sulphate solution and forms ferrous sulphate, which is light green: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$. As blue $CuSO_4$ is used up and $FeSO_4$ forms, the blue colour fades to light green, and brown copper is deposited on the nail.

Exam mein aise likho

Exercise Q14. In the refining of silver, the recovery of silver from silver nitrate solution involved displacement by copper metal. Write down the reaction involved.

$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$ (copper is more reactive than silver)

Ab tumhari baari *answers topic ke end mein*

1. Zinc granules are added to $CuSO_4$ solution. What change is seen in the colour of the solution and on the zinc surface?

   Answer

2. Complete the equation $Pb + CuCl_2 \rightarrow ?$ Which metal is more reactive?

   Answer

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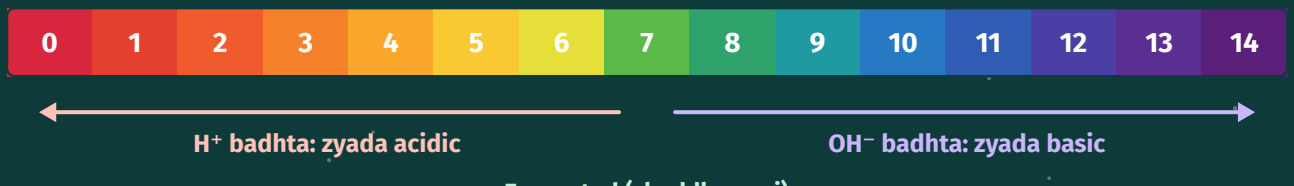
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Topic 5

pH scale aur rozana zindagi

Kitna acidic, kitna basic?

0 se 14: ek number mein poori kahani



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Zoya aur Kabeer ki kahani



Zoya
Sir, mere naye toothpaste par likha hai 'pH balanced', aur shampoo par bhi. Yeh pH kya cheez hai?

Kabeer
Aur coach kehte hain cold drink zyada piyoge to daant kharab honge. Kya isme bhi pH ka haath hai?



Adeel Sir

Adeel Sir
Haan! **pH** ek scale hai jo batata hai solution kitna acidic ya basic hai. Daant, pet, mitti aur baarish: sab pH se chalte hain. Aaj imli wala mystery bhi solve karenge!

Indicator sirf batata hai acid ya base; **universal indicator** aur **pH scale** batate hain **kitna**. pH 7 neutral, 7 se kam acidic, 7 se zyada basic. Hamara khoon, pet, daant, mitti aur baarish: sab ek khaas pH par hi theek kaam karte hain.

Is topic mein

pH scale, universal indicator, Activity 2.11 (Intext Q1 to Q3; Ex 12)	**24**
Strong vs weak; milk aur curd (Exercises 1, 9 to 11)	**25**
pH aur zindagi: pet, daant, mitti, acid rain (Intext Q4; Ex 4; Table 2.3)	**26**
Mystery solved: imli aur chamakte bartan	**27**
Ek nazar mein + quick check	**28**

Padhne ke baad tick karo

- ☐ pH scale aur universal indicator samjha sakta/sakti hoon
- ☐ Strong aur weak acids/bases ka farq bata sakta/sakti hoon
- ☐ Pet, daant, mitti aur acid rain mein pH ka role samjha sakta/sakti hoon
- ☐ Natural acids (Table 2.3) aur bee sting ka ilaaj bata sakta/sakti hoon

Adeel Sir

Adeel Sir kehte hain
pH jitna **kam**, H^+ utna **zyada**, aur solution utna **zyada acidic**. pH 1 wala, pH 4 wale se kahin zyada acidic hai.

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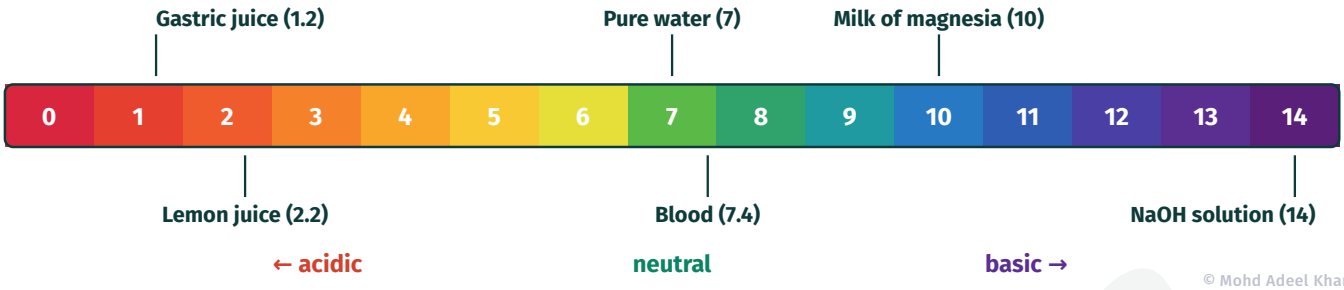
Section 2.3

Samajh aaya? Rate karo ☆ ☆ ☆

pH scale: acid-base ka thermometer

Adeel Sir ka example

Thermometer batata hai kitna **garam**, pH scale batata hai kitna **acidic ya basic**. pH mein 'p' German shabd *potenz* (power) se hai. Scale 0 (bahut acidic) se 14 (bahut basic) tak jaata hai; beech mein **7 = neutral**.



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Universal indicator: kai indicators ka mixture, jo H^+ ke alag-alag concentrations par **alag-alag colours** deta hai. Isse pH nikaalte hain. **Strong acids** (HCl) zyada H^+ dete hain, **weak acids** (acetic acid) kam. Isi tarah strong base NaOH zyada OH^- deta hai.

Activity 2.11: solution	Saliva (before meal)	Saliva (after meal)	Lemon juice	Aerated drink	Tomato juice	Tap water	1 M NaOH	1 M HCl
pH (approx)	7.4	5.8	2.2	3	4.2	7	14	0

Values approx hain; tumhare pH paper ka colour thoda alag aa sakta hai. Carrot juice aur coffee bhi halke acidic aate hain.

Exam mein aise likho

Exercise Q12. A milkman adds a very small amount of baking soda to fresh milk. (a) Why does he shift the pH of the fresh milk from 6 to slightly alkaline? (b) Why does this milk take a long time to set as curd?

(a) Baking soda is a mild base. Making milk slightly alkaline neutralises the acid formed, so the milk does not turn sour quickly and can be kept longer. (b) The lactic acid formed during curdling is first used up in neutralising the baking soda, so it takes longer for the milk to become acidic enough to set as curd.

Exam mein aise likho

Intext Q1 (page 28). You have two solutions, A and B. The pH of solution A is 6 and pH of solution B is 8. Which solution has more hydrogen ion concentration? Which of this is acidic and which one is basic?

Solution A (pH 6) has more hydrogen ion concentration, because lower pH means more H^+ ions. A is acidic (pH less than 7) and B is basic (pH more than 7).

Exam mein aise likho

Intext Q2. What effect does the concentration of H^+(aq) ions have on the nature of the solution?

The higher the concentration of H^+(aq) ions, the more acidic the solution (and the lower its pH).

Exam mein aise likho

Intext Q3. Do basic solutions also have H^+(aq) ions? If yes, then why are these basic?

Yes. But in basic solutions the concentration of OH^- ions is more than that of H^+ ions, so they are basic.

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Topic 3

Acids, salts aur reactivity series

Kaun kise hataayega?

Reactivity series: kaun kitna bechain?

K	Na	Ca	Mg	Al	Zn	Fe	Pb	H	Cu	Hg	Ag	Au

reactivity badhti hai (left ki taraf)

H se upar: dilute acid se H_2

upar wala, neeche wale ko salt se hataata hai

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Zoya aur Kabeer ki kahani



Zoya
Sir, maine pital ke glass mein nimbu-paani rakha to thodi der mein swaad **kasaila** ho gaya. Steel mein aisa kyun nahi hua?

Kabeer
Aur lab mein loha ki kil $CuSO_4$ mein daali to uspar **laal-bhoori parat** chadh gayi. Copper kahan se aaya?



Adeel Sir

Adeel Sir
Har metal ki **reactivity** alag hai. Acid kisse react karega, aur kaun sa metal doosre ko uske salt se **bhaga dega**: aaj iski poori list (reactivity series) banayenge.

Zyadatar metals dilute acids se react karke **salt aur hydrogen** dete hain. Ek zyada reactive metal kam reactive metal ko uske **salt solution se displace** kar deta hai. Metals ko reactivity ke order mein lagao to bani **reactivity (activity) series**.

Is topic mein

Activity 3.11: metals + dilute acid, aqua regia (Intext Q4, p46; Ex 7)	**15**
Activity 3.12: displacement (Intext Q3, Q5, p46; Ex 1, 4)	**16**
Table 3.2: reactivity series + practice	**17**
Ek nazar mein + quick check	**18**

Padhne ke baad tick karo

- ☐ Activity 3.11: Mg, Al, Zn, Fe ka order bata sakta/sakti hoon
- ☐ HNO_3 se H_2 kyun nahi milti, samjha sakta/sakti hoon
- ☐ Activity 3.12: displacement ke equations likh sakta/sakti hoon
- ☐ Reactivity series yaad hai aur usse predict kar sakta/sakti hoon

Adeel Sir

Adeel Sir kehte hain

Metal + dilute acid → Salt + H_2.
Metal A + salt of B → salt of A + Metal B (agar A, B se zyada reactive ho).

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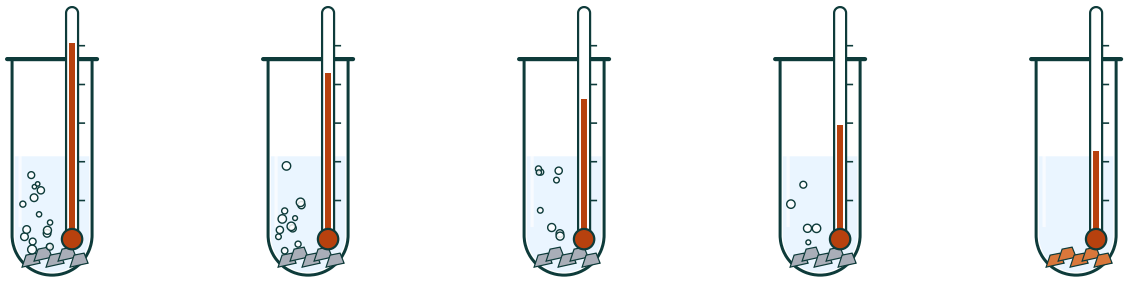
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Section 3.2.3

Samajh aaya? Rate karo ☆ ☆ ☆

Activity 3.11: acid mein kaun kitna tez?

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Mg + dil. HCl
bahut tez, sabse garam

Al + dil. HCl
tez

Zn + dil. HCl
theek-thaak

Fe + dil. HCl
dheere

Cu + dil. HCl
koi bubble nahi

NCERT Activity 3.11 · metals + dilute HCl

Kya chahiye
Mg, Al, Zn, Fe, Cu ke saaf tukde (Na, K nahi), dil. HCl, test tubes, thermometers

Kya karna hai
1. Har test tube mein 10 mL dil. HCl lo aur thermometer latkao.
2. Har tube mein ek metal daalo; bubbles ki speed aur temperature dekho.

Kya dikhta hai
Bubbles (H_2) aur temperature: **Mg > Al > Zn > Fe**. Mg sabse tez aur sabse garam. **Cu** mein koi reaction nahi.

$$Mg + 2HCl \rightarrow MgCl_2 + H_2 \;|\; 2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2 \;|\; Zn + 2HCl \rightarrow ZnCl_2 + H_2 \;|\; Fe + 2HCl \rightarrow FeCl_2 + H_2$$

Nitric acid ka twist: HNO_3 ek **strong oxidising agent** hai. Jo H_2 banti hai use woh **paani** mein badal deta hai, aur khud N ke oxides (N_2O, NO, NO_2) ban jaata hai. Isliye metals HNO_3 se H_2 nahi dete; sirf **Mg aur Mn** bahut dilute HNO_3 se H_2 dete hain.

Adeel Sir

Adeel Sir ka tip

Aqua regia ('royal water'): conc. HCl aur conc. HNO_3 ka **3 : 1** mixture, taaza banaya hua. Yeh **sona aur platinum** tak ghol deta hai, jo akele kisi acid mein nahi ghulte. Bahut corrosive, fuming liquid.

Clue 3 Sona dilute acids mein nahi ghulta, par **aqua regia** mein ghul jaata hai. Woh 'khaas ghol' kya tha?

Exam mein aise likho

Intext Q4 (page 46). Which gas is produced when dilute hydrochloric acid is added to a reactive metal? Write the chemical reaction when iron reacts with dilute H_2SO_4.

Hydrogen gas is produced. $Fe(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2(g)$

Exam mein aise likho

Exercise Q7. Name two metals which will displace hydrogen from dilute acids, and two metals which will not.

Metals above hydrogen in the reactivity series displace hydrogen from dilute acids, e.g., **zinc and magnesium** (also iron, aluminium). Metals below hydrogen do not, e.g., **copper and silver** (also gold).

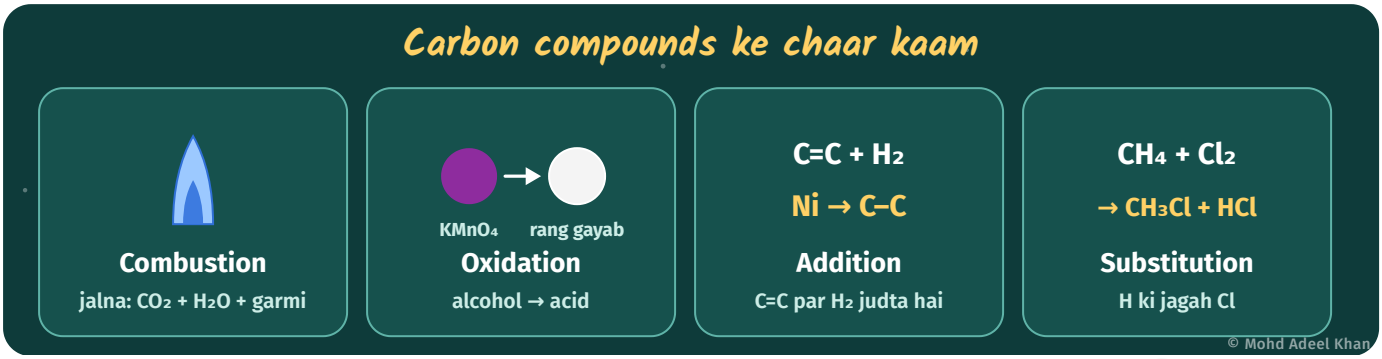
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Topic 4

Chemical properties

Jalna, oxidation, addition, substitution

Carbon compounds ke chaar kaam



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Zoya aur Kabeer ki kahani

Kabeer
Sir, mummy ki kadhai neeche se kaali kyun ho jaati hai? Gas ki flame to neeli hoti hai na?

Zoya
Aur dukaan wala bolta hai vanaspati ghee **tel se** banta hai. Tel jamkar ghee kaise ban gaya?

Adeel Sir
Dono ka answer carbon compounds ke **chaar reactions** mein hai: **combustion, oxidation, addition** aur **substitution**. Chalo ek ek karke dekhte hain.

Carbon compounds ka sabse bada kaam hai **fuel** banna: jalkar woh garmi aur roshni dete hain. Inke alawa alcohols **oxidise** hokar acids bante hain, unsaturated compounds mein **addition** hota hai aur saturated compounds mein **substitution**. Yahi chaar reactions board exam ke favourite hain.

Is topic mein

Combustion, Activity 4.3 (Ex 3, 9)	**23**
Activity 4.4: flame aur air hole (Intext Q2, p71)	**24**
Activity 4.5: oxidation (Intext Q1, p71; Intext Q2, p74)	**25**
Addition: hydrogenation (Ex 12, 13)	**26**
Substitution + test (Ex 14)	**27**
Ek nazar mein + quick check	**28**

Padhne ke baad tick karo

- ☐ Combustion ke equations aur flame ka rang samjha sakta/sakti hoon
- ☐ Activities 4.3, 4.4, 4.5 ke observations bata sakta/sakti hoon
- ☐ Hydrogenation aur uska industrial use likh sakta/sakti hoon
- ☐ Saturated aur unsaturated ko test se pehchaan sakta/sakti hoon

Adeel Sir

Adeel Sir kehte hain
Saturated compound: saaf neeli flame, substitution. **Unsaturated**: peeli dhuen wali flame, addition.

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Section 4.3.1

Samajh aaya? Rate karo ☆ ☆ ☆

Combustion: carbon compounds ka jalna

Carbon aur uske almost sabhi compounds oxygen mein jalkar **CO_2** (aur H ho to **H_2O**) dete hain, saath mein bahut **garmi aur roshni**:

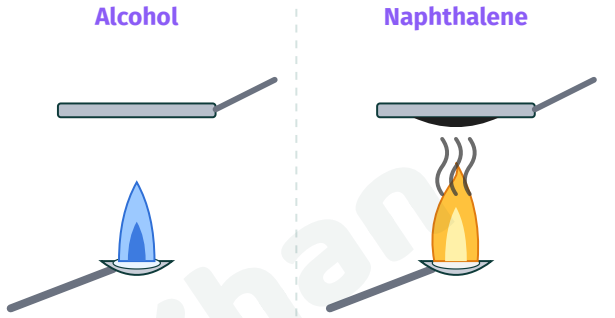
$$C + O_2 \rightarrow CO_2 + \text{heat and light}$$
$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat and light}$$
$$CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O + \text{heat and light}$$

NCERT Activity 4.3 · Flame kaisi hai, dhuan hai?

Kya chahiye
Naphthalene, camphor, alcohol, spatula, metal plate

Kya karna hai
1. Ek ek compound spatula par lekar jalao.
2. Flame ka rang aur dhuan dekho.
3. Flame ke upar metal plate pakdo: kis par kaala parat jamti hai?

Kya dikhta hai
Alcohol **saaf neeli** flame se jalta hai, plate saaf rehti hai. Naphthalene **peeli flame** aur kaale dhuen se jalta hai: plate par **soot (kaala carbon)** jam jaata hai. Camphor ka result khud note karo.



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Nateeja: saturated hydrocarbons aam taur par **saaf flame** dete hain, jabki unsaturated compounds **peeli flame aur bahut kaala dhuan** dete hain. Par saturated hydrocarbons bhi **kam hawa** mein jalein to dhuen wali flame deti hain.

Adeel Sir ka solved example 4.1

Exercise Q3. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that (a) the food is not cooked completely (b) the fuel is not burning completely (c) the fuel is wet (d) the fuel is burning completely.

Socho Kaala parat = unburnt carbon (soot)

Kyun Hawa kam: fuel poora nahi jal raha (stove ke air holes band)

Answer: (b) the fuel is not burning completely

Exam mein aise likho

Exercise Q9. Why are carbon and its compounds used as fuels for most applications?

Carbon and its compounds burn in air or oxygen to give carbon dioxide and water, releasing a large amount of heat and light. They have a high calorific value, burn in a controllable way, and are easy to store and transport. Hence wood, coal, LPG, CNG, petrol and kerosene are used as fuels.

Adeel Sir

Adeel Sir ka tip

Equation balance karne ka shortcut: pehle **C** (CO_2 se), phir **H** (H_2O se), sabse aakhir mein **O**.

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Reaction map: poora chapter ek page par

Acid + metal
Salt + H_2
$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
H_2: pop test
2 page 9

Acid + carbonate
Salt + CO_2 + H_2O
$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
CO_2: lime water milky
2 page 9

Acid + base
Salt + H_2O
$NaOH + HCl \rightarrow NaCl + H_2O$
Neutralisation
3 page 14

Acid + metal oxide
Salt + H_2O
$CuO + 2HCl \rightarrow CuCl_2 + H_2O$
Metal oxides basic
3 page 14

Base + non-metal oxide
Salt + H_2O
$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
Non-metal oxides acidic
3 page 14

Indicator	Acid mein	Base mein	Yaad rakhne ki baat
Litmus	Blue → red	Red → blue	Lichen se; BAR / RBB
Phenolphthalein	Colourless	Pink	Neutralisation activity
Methyl orange	Red	Yellow	Synthetic
Onion, vanilla	Smell rehti hai	Smell jaati hai	Olfactory

Salt ka pH predict karo

	Strong base	Weak base
Strong acid	**pH = 7: neutral** NaCl, KNO_3, Na_2SO_4	**pH < 7: acidic** NH_4Cl, $CuSO_4$, $AlCl_3$
Weak acid	**pH > 7: basic** Na_2CO_3, $NaHCO_3$, CH_3COONa	**depends (extra)** $CH_3COONH_4 \approx 7$

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pH ki numbers	Value
Body ki range	7.0 to 7.8
Acid rain	below 5.6
Tooth decay	below 5.5
Neutral (pure water)	7

⚠ **Galti mat karna.** **Dilute** ka matlab **weak** nahi. Strong/weak = kitna ionise hota hai (H^+ dena); dilute/concentrated = kitna paani mila hai. Dilute HCl bhi **strong acid** hai.

Exam mein aise likho

Define in one line each: acid, base, alkali, salt, neutralisation, pH.

Acid: a substance that gives $H^+(aq)$ [H_3O^+] ions in water. **Base:** a substance that gives $OH^-(aq)$ ions in water. **Alkali:** a base that dissolves in water. **Salt:** a compound formed, along with water, when an acid reacts with a base. **Neutralisation:** the reaction of an acid with a base to form salt and water. **pH:** a scale (0 to 14) that measures the hydrogen ion concentration of a solution; lower pH means more acidic.

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Board pattern test: Section A

Adeel Sir ke test rules

Book band karo. Poore test ke liye **90 minutes**, total **40 marks** (CBSE Class 10 pattern). Section A: har question **1 mark**, sirf ek option.

Section A: ____ / 12

Q1 [Based on CBSE 2025] During the preparation of dry HCl gas, calcium chloride in the guard tube is used to:

◯ absorb HCl gas ◯ absorb moisture
◯ produce chlorine ◯ neutralise the acid

Q2 Which of the following turns phenolphthalein pink?

◯ lemon juice ◯ vinegar
◯ soap solution ◯ common salt solution

Q3 The pH of the gastric juice released during digestion is:

◯ less than 7 ◯ more than 7
◯ equal to 7 ◯ equal to 0

Q4 Which of the following salts gives a basic solution in water?

◯ NaCl ◯ KNO_3
◯ Na_2CO_3 ◯ NH_4Cl

Q5 When zinc is warmed with sodium hydroxide solution, the products are:

◯ zinc hydroxide and oxygen
◯ sodium zincate and hydrogen
◯ zinc oxide and water
◯ no reaction

Q6 Which of the following is NOT a use of baking soda?

◯ antacid
◯ baking powder
◯ soda-acid fire extinguisher
◯ removing permanent hardness

Q7 Plaster of Paris is obtained by heating:

◯ limestone ◯ gypsum
◯ washing soda ◯ bleaching powder

Q8 Which of the following has the highest pH?

◯ lemon juice ◯ pure water
◯ blood ◯ milk of magnesia

Q9 In the chlor-alkali process, chlorine gas is obtained at the:

◯ cathode ◯ anode
◯ both electrodes ◯ bottom of the cell

Q10 Bleaching powder is used for:

◯ softening hard water
◯ disinfecting drinking water
◯ making cakes
◯ setting bones

Assertion-Reason *(Q11 and Q12, 1 mark each)*

(a) Both A and R are true and R is the correct explanation of A. **(b)** Both A and R are true but R is not the correct explanation of A. **(c)** A is true but R is false. **(d)** A is false but R is true.

Q11 **Assertion (A):** [Based on CBSE 2025] For dilution, water is added slowly to concentrated nitric acid.
Reason (R): The process of dissolving an acid in water is highly exothermic.

◯ (a) ◯ (b) ◯ (c) ◯ (d)

Q12 **Assertion (A):** A solution of ammonium chloride in water is acidic.
Reason (R): Ammonium chloride is a salt of a strong acid and a weak base.

◯ (a) ◯ (b) ◯ (c) ◯ (d)

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Poori book chahiye?

Is sample mein har chapter ke sirf do pages the. Poori **Chemophile Foundation 2 (for 10th Class)** mein 4 chapters ke **210 pages** hain, har topic isi style mein.

Adeel Sir

Copies aur jaankari ke liye

Mohd Adeel Khan

MSc BEd (JMI)

Ph. 9953821875

Students ke liye

- ✓ Concept pehle, ratna baad mein
- ✓ NCERT ka har question solved
- ✓ Board ke liye English answers
- ✓ Har chapter ke end mein 40 marks ka test

Teachers aur schools ke liye

- ✓ NCERT section numbers ke saath
- ✓ Har topic ke ready-made visuals
- ✓ Recap, quick check aur answer key
- ✓ Revision: fact ya formula sheet, top mistakes

Is sample mein kaun se topics the

Chapter	Sample topic	Poora chapter
1. Chemical Reactions and Equations	Topic 5: Displacement aur double displacement	54 pages
2. Acids, Bases and Salts	Topic 5: pH scale aur rozana zindagi	49 pages
3. Metals and Non-metals	Topic 3: Acids, salt solutions aur reactivity series	51 pages
4. Carbon and its Compounds	Topic 4: Carbon compounds ki chemical properties	56 pages

Saath mein extra

Periodic Classification bonus booklet (29 pages), Quick Revision Sheets (har chapter ek page), 2 full Mock Test Papers aur sab kuch ek saath Complete Volume (250 pages) mein bhi.

Chemophile Foundation 1 (for 9th Class) bhi ready hai

3 chapters, 121 pages: usi style mein, usi author ki taraf se.

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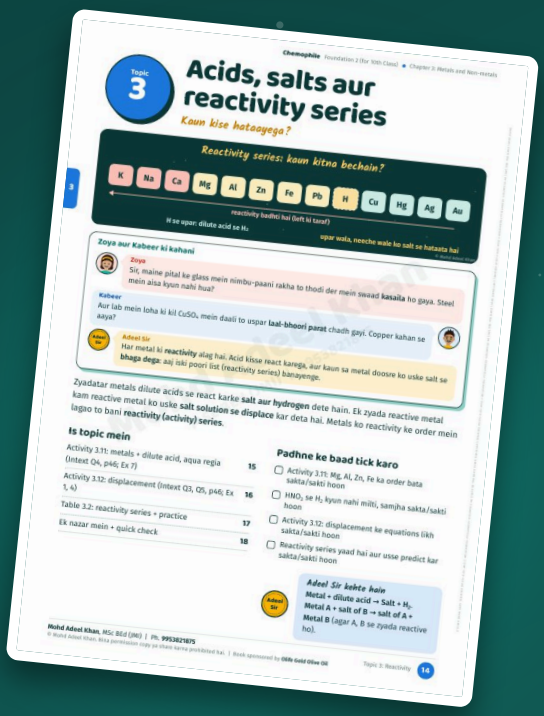
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Chemophile

Foundation 2 (for 10th Class)

Free sample

Chemistry ratni nahi, samajhni hai.



4 chapters

210 colourful pages

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Poori book ke liye contact karein: **Mohd Adeel Khan**, MSc BEd (JMI)

Ph. **9953821875**

Author

Mohd Adeel Khan

MSc BEd (JMI)

Ph. **9953821875**

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